



Use of Balanced Scorecards for Knowledge Management Roadmap

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Knowledge Management (KM) is a new scientific discipline which facilitates organisations to better perform in their highly competitive environment. Universities are learning organisations in the centre of the knowledge triangle. They are faced with the need to better manage their knowledge resources and to facilitate their transfer to all interested stakeholders. Similar to companies, universities are developing at different speeds, and some are lagging behind in adopting KM strategic approach. The goal of this paper is to provide guidance to university staff in developing their KM strategy roadmap on bases of the Balanced Scorecard approach and integrating it in the overall strategy.

Categories and Subject Descriptors: **K.4 [Computers and Society: Organisational Impacts; Electronic Commerce; K.6 [Management of Computing and Information Systems]: Project and People Management; Systems analysis and design; J.4 [Social and Behavioural Sciences]: Psychology,**

General Terms: Knowledge Management Strategy

Additional Key Words and Phrases: Knowledge Management, Organisational Patterns, Universities

ACM Reference Format:

Gourova, E., Dragomirova, M. and Zografova, L. 2014. Use of Balanced Scorecards for Knowledge Management Roadmap. EuroPLoP 2014, 7 pages.

1. INTRODUCTION

Nowadays organisations and individuals world-wide face enormous changes in technologies, in the ways of doing business, in work and life. In order to remain competitive and to keep their position on the market or job, both organisations and individuals need to adapt fast to the changes in their environment. This calls for a short learning cycle of individuals and organisations. It requires better management of all knowledge resources of the organisation in order to allow employees to easily find the knowledge they need in order to learn, to generate new knowledge or apply it in their everyday tasks. The company practice has shown that knowledge, when properly used and leveraged, could drive organisations to become more innovative and thus, more, competitive [2].

The adoption of Knowledge Management (KM) as company practice was driven initially by large organisations and consulting companies world-wide [3]. The concepts, tools and methods of KM are recognised to be important for all knowledge-intensive organisations, both in the public and private sectors [3]. Improving decision making and knowledge sharing, faster innovation, reducing duplication of work and improving business processes, and managing effectively knowledge resources could constitute important reasons for knowledge-based organisations to undertake a KM initiative. However, the expected KM benefits depend on the proper KM implementation and the following monitoring.

The aim of this paper is to provide guidance for preparing a KM strategic plan and corresponding metrics needed for monitoring and control. The strategy development process follows the Balanced Scorecard (BSC) approach of Kaplan and Norton [10], which considers four strategic perspectives:

- financial perspective: aimed at higher revenue and higher return of investments;

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EuroPLoP 2014, July 9 - 12 2014, Irsee, Germany Copyright 2014 ACM ...\$15.00

<http://dx.doi.org/10.1145/2721956.2721975>

- customers perspective: focused on customers relations, customers satisfaction with products or services attributes, and organisational image;
- internal processes perspective: includes operations management for production or delivery of services; customers management; innovation; and regulatory and social processes.
- learning and growth perspective: includes human capital, information capital and organisational capital.

These perspectives are interrelated: The active and synchronised use of the intellectual capital of the organisation enhances the internal process efficiency which generates better satisfaction of customers, and subsequently, improves financial results [10].

As suggested by Keyes [11], the implementation of BSC in KM should focus on future readiness instead of learning and growth. In this case, in each perspective of the BSC should be linked the goals set to specific targets to be met applying concrete activities (Figure 1). The development towards the targets should be measured with specific key performance indicators (KPI).

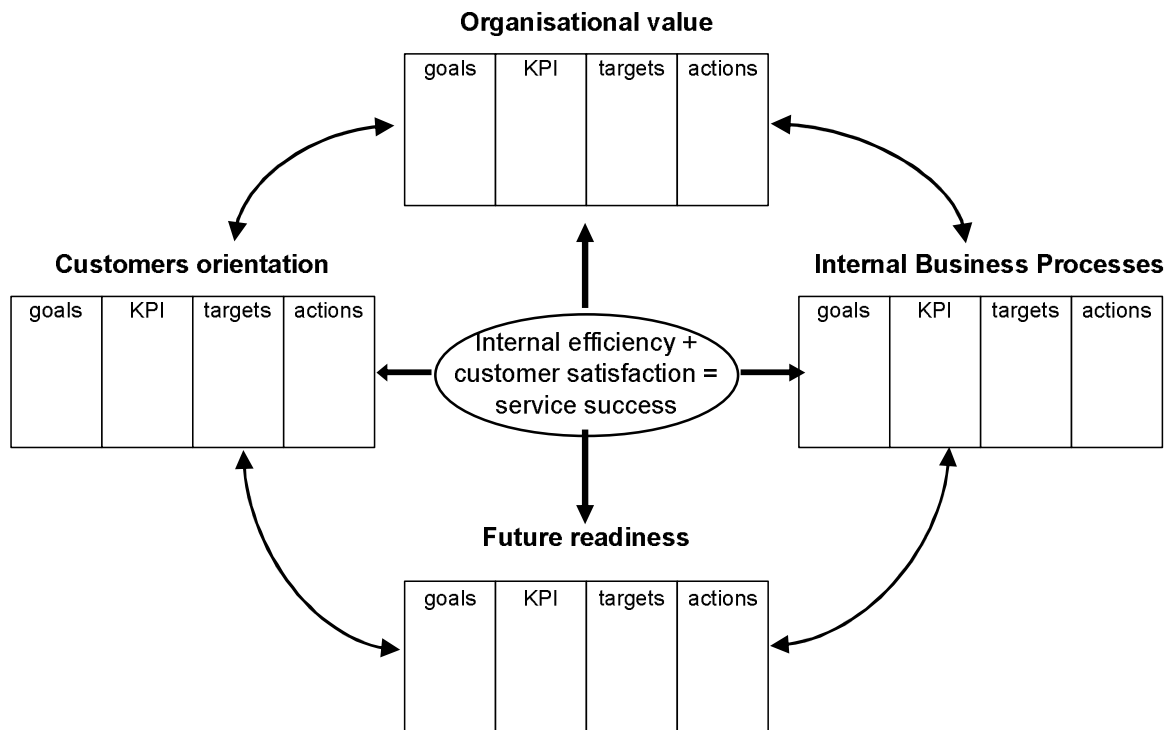


Figure 1 Use of BSC for KM, adapted [11]

The paper logically follows the results of the Knowledge Audit (KA), carried out according to a set of patterns presented at workshops of EuroPLoP 2009, 2010, 2011 and 2012 [4, 5, 6, 7], and the initial patterns for KM system design presented at VikingPLoP 2014 [8] and on KM strategy development - at EuroPLoP 2013 [9].

These patterns are intended to codify business practices in the area of Knowledge Management in a pattern language so that they may be better understood, communicated, applied and studied. The previous work of the authors focused on Knowledge Audit tools used, and started with a set of patterns on Knowledge Management System (KMS) development according to the patterns on Knowledge Management strategy. The latter is further elaborated in this paper. Figure 2 shows the relations between the different patterns.

The STRATEGIC ROADMAP could include different perspectives to which the organisation should pay attention by developing its KM strategy. Some sub-patterns will be also considered linked to specific

PERFORMANCE MEASUREMENT. The concrete shape of the STRATEGIC ROADMAP depends on the implementation alternative chosen (KNOWLEDGE MANAGEMENT STRATEGIC CHOICE [9]) derived from the mission of the KM initiative in the organisation (KNOWLEDGE MANAGEMENT MISSION [9]), and the goals set (KNOWLEDGE MANAGEMENT GOALS [9]). The KNOWLEDGE AUDIT REPORT [4] provides details regarding the current state in the organisation which can be used as input data for the KMS design. Each of the tools of the KNOWLEDGE AUDIT METHODOLOGY [4] provides different snap-shots on the organisational state, and builds a base for strategic decisions making and detailed planning.

The patterns are intended for Knowledge Management practitioners, teachers, and students. The patterns may be applied in the context of knowledge-intensive public or private organisations.

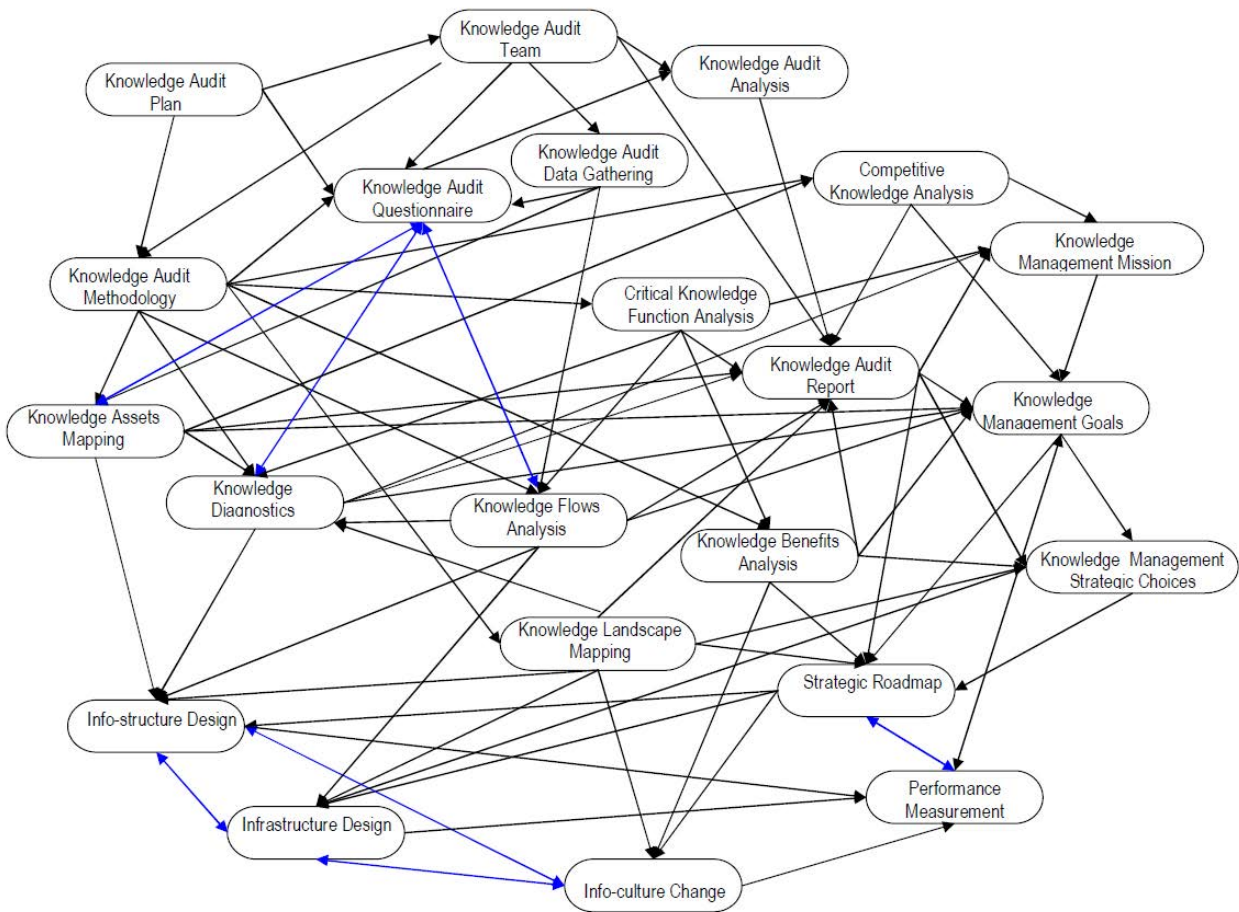


Figure 2 Patterns overview [8]

2. STRATEGIC ROADMAP

A university needs to improve its performance as knowledge-intensive organisation, and better manage the knowledge of its researchers. It has an organisational strategy, and has determined the main goals and the strategic alternative for KM implementation. It needs to design its knowledge management system and to provide clear guidance to all employees for KM implementation.

2.1. Problem: How to turn goals and priorities in a concrete action plan for KM implementation?

The limited resources face the organisation with a need to determine where to invest. Some actions require immediate reaction, while others can be postponed. KM implementation requires additional

resources, both human and financial. However, it provides opportunities for increasing the efficiency and the quality of organisational working processes which pays off in the longer run.

Knowledge is acquired from internal and external sources. Supporting sharing of internal knowledge can facilitate the performance of knowledge workers and the organisation as a whole. At the same time, in the globalised world top-level research cannot be carried out in house and requires exchange of knowledge with external stakeholders.

Research responds to economic and societal needs. Monitoring the rapid changing external environment can provide essential knowledge for better responding to stakeholders needs and the emerging trends in economy and society.

Responding to on-going processes in their environment, and fast adapting to the services demanded by stakeholders is a need for all organisations nowadays in order to remain competitive. However, academic institutions are characterised in most cases by a conservative nature, lack of flexibility and willingness to change.

Research processes are knowledge-intensive business processes with specific context and needs for knowledge resources. Their efficiency depends on the flows of qualitative knowledge, both tacit and explicit, coming from internal and external sources.

Various sophisticated software products are available and can facilitate the knowledge processes. Technology alone would not make KM work. The availability of content easy-to-find, easy-to-use, and up-to-date is essential for knowledge workers.

Academic organisations take measures to build knowledge bases and provide access to the explicit knowledge stored there. However, the employees are owners, users and providers of knowledge resources. Gathering the tacit knowledge and expertise existing in the organisation and in its environment depends on the willingness of people to share it, as well as on their skills to process it.

Knowledge creation and transfer are in the centre of academic institutions working processes. However, the ownership of new intellectual assets is not always clear, and knowledge workers, if not properly motivated, are not eager to share their results.

2.2. Solution: Use the Balanced Score Card as a base for planning, and focus on ensuring proper knowledge flows to business processes in place.

The already set KM mission, KM goals and the strategic alternative chosen [9] provide a framework for determining concrete actions for achieving the goals set.

According to the available funding prioritise the KM activities:

- Consider the expected KM benefits identified during the Knowledge Audit (e.g. KNOWLEDGE MANAGEMENT BENEFITS ASSESSMENT [6]).
- Start with activities which will bring visible results faster and will meet the needs of knowledge workers.
- Ensure the necessary resources for the actions planned.

Take into account the activities dependencies when planning the time schedule:

- Start with the changes in Learning and Growth perspective according to INFO-CULTURE CHANGE [9].
- Make changes in Internal Processes perspective according to INFRASTRUCTURE DESIGN and INFO-STRUCTURE DESIGN [9].

Make clear the responsibilities for all activities planned. For monitoring purposes specify the expected results and set assessment criteria.

Choose the KM tools for support of knowledge processes at individual, groups and organisational level according to the analysis made during the Knowledge Audit, and more specifically of the business processes, knowledge processes and knowledge flows [7]:

- When building knowledge bases take into account the needs and requirements of knowledge workers and balance them with organisational needs.
- Ensure diversity of knowledge flows from external and internal sources.
- Balance the use of information technologies with techniques which facilitate collaboration and knowledge sharing.

Ensure cohesion of all activities undertaken in each of the BSC perspectives. Do not forget that the activities set for INFO-CULTURE CHANGE [9] have impact on the proper implementation of all activities focused on changes in internal processes, and influence the customers orientation and the organisational value.

2.3. Consequences

If the changes in organisational culture and human resources policy (learning and growth) are not made early enough, any changes in the knowledge and business processes could fail.

Meeting the needs of employees, ensuring their motivation and commitment facilitates the changes in internal processes and the implementation of all KM activities.

If not linked to employees needs, knowledge and working processes in place, even the most advanced technology cannot deliver the expected benefits.

Ensuring resources for KM implementation once is not sufficient – it requires continuous efforts.

Without sufficient financial resources, and carefully prioritised actions the KM-related changes in the organisation could hardly be fulfilled.

Providing employees fast and easy access to the knowledge they need facilitates their work and ensures better service to customers.

The rapid changes outside organisations could face it with risks of failing to achieve its goals if the external knowledge is not timely available to employees.

Ensuring synergy of organisational, technological and human activities is a factor for meeting the expected KM benefits and return of the investments made.

Monitoring and control can add overhead to the KM implementation, but early warning helps overcoming bigger financial losses, and ensuring higher quality of the internal processes.

2.4. Example

The strategy modeling process at the English Language Faculty of Engineering (ELFE) of the Technical University – Sofia (TU) used business process management tools and created a BSC system as a value-added chain process. The most important reason for using the BSC approach was that it easily links the objectives set with related measures, targets and key performance indicators for monitoring of the implementation results.

Table 1 ELFE strategy map - Learning and growth perspective

Goals	Activities
1.1. Developing a vision for future research	▪ desk research in the area ▪ users' needs assessment ▪ identification of interested Bulgarian stakeholders ▪ exchange of knowledge with partners ▪ setting collaborative research roadmaps
1.2. Strengthening human resources	▪ reintegration of researchers ▪ hiring young researchers ▪ career plan development ▪ research supervision/ mentoring
1.3. Enhancing knowledge in the research field	▪ targeted research workshops

	▪ learning-through-research ▪ mobility visits to partners ▪ internship in industry ▪ grants for participation at external events
1.4. Enhancing IPR, innovation and management capacity	▪ training on IP and technology transfer (TT) ▪ training on innovation management ▪ training on research communication and industry collaboration

For the purposes of the ELFE KM strategy was developed a strategy map consisting of three perspectives - Learning and growth (Table 1), Internal processes (Table 2) and Stakeholders (Table 3). For each perspective are determined specific strategic goals, that are related to each other in a hierarchy and the main goal is, thus, broken down into sub-goals. As ELFE is a knowledge-based publicly funded organisation, the financial perspective was not elaborated.

Table 2 ELFE strategy map - Internal processes perspective

Goals	Activities
2.1. Improving research environment	▪ new computer places for researchers ▪ update of specialised labs ▪ access to engineering scientific books and journals
2.2. Improving knowledge management processes	▪ development of collaborative platform ▪ development of on-line research databases and open library ▪ development of TU yellow pages ▪ development of stakeholders database ▪ integrating Web 2.0 tools
2.3. Improving quality of research	▪ Joint supervision/mentoring by experienced researchers ▪ Monitoring of the quality of research results and publications
2.4. Improving IP and knowledge management	▪ preparation of IP and knowledge transfer (KT) Plan ▪ inventory of ELFE knowledge assets ▪ valorisation of research results

Table 3 ELFE strategy map - Stakeholders perspective

Goals	Activities
3.1. Enhancing international networking in research	▪ twinning with EU research partners ▪ organising brokerage events ▪ launching joint projects ▪ organising joint events
3.2. Enhancing networking with industry	▪ building national research clusters ▪ launching joint projects ▪ improving knowledge and technology transfer ▪ organising joint events
3.3. Increasing the TU visibility	▪ Organising highly visible events ▪ Improving web presence ▪ Dissemination of ELFE materials
3.4. Increasing the image of researchers in the society	▪ Organising open TU days ▪ Undertaking outreach activities ▪ Targeted communications over printed and electronic media ▪ Dissemination of leaflets

ACKNOWLEDGEMENTS

The authors express their deepest appreciation to their shepherd Dietmar Schuetz for the thorough guidance for preparing this paper, and to all members of the workshop at EuroPLoP 2014.

The authors gratefully acknowledge the financial support provided for attending the conference under the Operational Programme Human Resources Development grant agreement BG051PO001-3.3.06.-0052

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